

Molecular Dynamics Simulation of Water by TIP5P Model

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<http://www1.m4.mediacat.ne.jp/dphysique/>

Procedures of molecular dynamics simulation for water and ice are written in the five-point water TIP5P model. Before, the shake/rattle model was used to show the ice not melting by microwaves - theory discovery in JCP 2007. Today, five-body water molecules are used with two hydrogens H1, H2 and two dummy sites L1, L2. The fifth oxygen site is assigned as Lennard-Jones potential $\Psi = A/r^{12} - B/r^6$. One then gets the water TIP5P model by the six-membered water or ice crystals.

- 1. "Classical Mechanics", H. Goldstein, C. Poole, J. Safko, 3rd Edition, Pearson Education Inc., England, 2003.*
- 2. "Microwave heating of water, ice and saline solution: Molecular dynamics study", M. Tanaka and M. Sato, J. Chem. Phys., 126, 034509 1-9 (2007).*
- 3. "Water and hydrate by molecular dynamics TIP5P simulation", M. Tanaka, <https://github.com/Mtanaka77/>.*

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***** May, 2023 ****
*
*  ## Molecular Dynamics of Water and Ice by TIP5P Code ##
*  - Microwave heating, ice below T=273 K is not melted
*
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*
*  Released by GPL-3.0 License, https://github.com/Mtanaka77/
*  Copyright (C) 2006-2024. All rights reserved.
*
*  Reference
*  1) M.Tanaka, J.Comput.Phys., vol. 79, 206 (1988).
*  2) M.Tanaka, J.Comput.Phys., vol.107, 124 (1993).
*  3) M.Tanaka, Comput.Phys.Comm., vol.87, 117 (1995).
*  4) M.Tanaka and M.Sato, J.Chem.Phys., 126, 034509 (2007).
*  5) M.Tanaka, Comput.Phys.Comm., vol.241, 56 (2019).
*
* -----
*
*  Files for this simulation
*  @p3mtip5p code name (p3m + tip5p)
*  07a is a run name and sequential number (a,b,c...)
*
*  1. @p3mtip5p07a.f03 : MD simulation code
*  2. param_tip5p_D07a.h : parameter file, physical constants
*  3. TIP507_config.start0 : parameter file, kstart=0
*  or continuation TIP507_config.start1 : kstart=1 or 3
*  4. Initial molecules (exyz and quaternion)
*  1cx666a.exyz/1cx666a.q for liquid water, 1cx666b.exyz in
*  230 K or mh3.exyz,mh3.q for methane hydrate.
*  Refer to if_xyz1 or if_xyz2 parts in subroutine /init/.
*
*  Histories:
*  Translation and rotation of molecules
*  5-point hydrogen and oxygen pairs
*  prefactor (realteil) and pref_eps (Lennard-Jones)
*  epslj_A,B for water, ep(i) for hybrid molecules.
*
*  Fujitsu FX100 by Feb.2020, NEC-Aurora from July 2020.
*
* ***** First code: 02/26/2005 *****
*
*  1. >>> run's name is given by param_tip5p_D07a.h
*
*  2. >>> Run parameters are given in TIP07_config.start0, or 1
*  which is read by /read_conf/.
*
*  3. >>> Start, Restart and continue runs.
*
*  Physical units:
*  t_unit= 0.0100d-12      ! 0.01 ps
*  a_unit= 1.0000d-08      ! 1 Ang
*  w_unit= 1.6605d-24*18.  ! H2O is the unit
*  e_unit= 4.8033d-10      ! esu
*
*  ^ dv      t^2 e^2  qq'  12 t^2 eps  r0      1 r0
*  m ---- = ---- + ---- + ---- [ (---)^12 - (---)^6 ]
*  dt      ma^3    r^2    ma^2  r      r      2 r

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Molecular dynamics simulation of water and ice molecules by TIP5P model

Type: Fortran 2003

Length 4000 lines

Main subroutines:

es3d_tip5, run_md, moldyn, realteil, forces_5, p3m_perform, perform_aliasing_sums, calculate_differential_operator, ..., sinc, init

Post-processing routine & graphics:

data write/read by fort.11, fort.12, fort.08, fort.13, fort.18, fort.77.ps

Download from the site:

https://github.com/Mtanaka77/Water_and_Hydrate_by_Molecular_Dynamics_TIP5P_Simulation/

Molecular Dynamics Simulation by the TIP5P Model

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Summary

- a. Five sites are oxygen (O), hydrogens (H_1, H_2), and the virtual sites (L_1, L_2). They have charges of 0.0, 0.241e, -0.241e, respectively. The L_1, L_2 are geometrically known from H-O-H and called the dummy sites.*
- b. Position $\mathbf{R}_j$ and velocity $\mathbf{V}_j$ are for the translation of $j = 1-N/5$ molecules, and that of $\mathbf{r}_i$ are for the rotation of $i = 1-N$ atoms. Rotation coordinate (x_i, y_i, z_i) means for the five sites. Separation of traslation and rotation coordinates is done at the starting step.*
- c. The half time step is first executed for a predictor, and the full step is made for a corrector.*
- d. Before the end of one step, the forces are calculated at $\mathbf{r}_i = \mathbf{R}_j + (A11 * xr(i) + A21 * yr(i) + A31 * zr(i))$, with (xr, yr, zr) being the rotation of principal coordinates, and the three sites out of $i = 1-N$, and the L sites are calculated by algebraic operation.*
- e. After correction of quaternions, go to the beginning of the cycle. The leap-frog method is used for the plasmas and waters.*

Each step of the code

* Each step consists of translation (1), rotation (2)-(4), and addition of field (5)-(8).

0. Read the quaternions from the file, read(30) e_0, e_1, e_2, e_3 , $j=1-N/5$.

1. Sum the sites, $d\mathbf{N}_j/dt = \sum_{k=1}^5 \mathbf{F}_{i,k}/m_j$, $d\mathbf{R}_j/dt = \mathbf{V}_j$ with the translational motion.

The (x_r, y_r, z_r) and (Im_x, Im_y, Im_z) of molecules are the constants of time.

2. $d\mathbf{L}_j/dt = \sum_{k=1}^5 (y_k F_{i,k}^z - z_k F_{i,k}^y, z_k F_{i,k}^x - x_k F_{i,k}^z, x_k F_{i,k}^y - y_k F_{i,k}^x)$ for the rotation matrix, summation over five sites.

3. $\omega_{j,\alpha} = (A_{\alpha 1}L_x + A_{\alpha 2}L_y + A_{\alpha 3}L_z)/Im_\alpha$ for species $A_{\alpha\beta}$ and moment inertia Im_α for $\alpha = x, y, z$.

4. $d\mathbf{q}_j/dt = (\Delta t/2)(-e_1\omega_x - e_2\omega_y - e_3\omega_z, e_0\omega_x - e_3\omega_y + e_2\omega_z, e_3\omega_x + e_0\omega_y - e_1\omega_z, -e_2\omega_x + e_1\omega_y + e_0\omega_z)$ for $\mathbf{q}_j$, $\boldsymbol{\omega}$, (e_0, e_1, e_2, e_3) by the Goldstein's book, and get a new rotation matrix.

5. Get a new rotation matrix $A_{\alpha,\beta}(e_0, e_1, e_2, e_3)$ of the time step with $\alpha, \beta = x, y, z$.

6. The rotation $\mathbf{r}_i$ and $(x_{r_i}, y_{r_i}, z_{r_i})$ are:

$$\mathbf{r}_i = \mathbf{R}_j + \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix} \begin{pmatrix} x_{r_i} \\ y_{r_i} \\ z_{r_i} \end{pmatrix}$$

at three sites $\mathbf{r}_i$ plus the virtual sites ($i = 1-N$), and $\mathbf{R}_j$ from Nr.1.

7. *Coulomb and LJ forces of the four sites and the oxygen site are calculated, which are very heavily consumed of CPU.*
8. *Corrections of quaternions at every 10 steps are executed, and go to the new time step.*

Quaternions in place of three-dimensional angles

$$e_0 = \cos \frac{\theta}{2} \cos \frac{\phi + \psi}{2}$$

$$e_1 = \sin \frac{\theta}{2} \cos \frac{\phi - \psi}{2}$$

$$e_2 = \sin \frac{\theta}{2} \sin \frac{\phi - \psi}{2}$$

$$e_3 = \cos \frac{\theta}{2} \sin \frac{\phi + \psi}{2}$$

*Classical Mechanics (3rd Edition)
H. Goldstein , C.P. Poole, J.Safko,
Pearson Education Inc., England 2003.*

Only three of them are independent
to avoid a gimbal lock

Quaternion representation (4.47)

Rotation matrix

$$\mathbf{r} = \mathbf{R} + \mathbf{A}^t \mathbf{r}'''$$

$$\mathbf{A} = \begin{pmatrix} e_0^2 + e_1^2 - e_2^2 - e_3^2 & 2(e_1 e_2 + e_0 e_3) & 2(e_1 e_3 - e_0 e_2) \\ 2(e_1 e_2 - e_0 e_3) & e_0^2 - e_1^2 + e_2^2 - e_3^2 & 2(e_2 e_3 + e_0 e_1) \\ 2(e_1 e_3 + e_0 e_2) & 2(e_2 e_3 - e_0 e_1) & e_0^2 - e_1^2 - e_2^2 + e_3^2 \end{pmatrix}$$

**Time derivative of
quaternions
e0,e1,e2,e3**

$$\begin{pmatrix} \dot{e}_0 \\ \dot{e}_1 \\ \dot{e}_2 \\ \dot{e}_3 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -e_1 & -e_2 & -e_3 & e_0 \\ e_0 & -e_3 & e_2 & e_1 \\ e_3 & e_0 & -e_1 & e_2 \\ -e_2 & +e_1 & e_0 & e_3 \end{pmatrix} \begin{pmatrix} \omega_{x'} \\ \omega_{y'} \\ \omega_{z'} \\ 0 \end{pmatrix}$$

In Goldstein's book, these definitions are sign reversed in notations, minus or plus signs !

$$(e_0, e_1, e_2, e_3) \Leftrightarrow (\chi, \eta, \xi, \zeta)$$

Goldstein

Ueda (Japan)

$$2\dot{\xi} = 2 \left(\sin \frac{\theta}{2} \sin \frac{\phi - \psi}{2} \right)'$$

$$= \dot{\theta} \cos \frac{\theta}{2} \sin \frac{\phi - \psi}{2} + (\dot{\phi} - \dot{\psi}) \sin \frac{\theta}{2} \cos \frac{\phi - \psi}{2}$$

$$= (\omega_x \cos \psi - \omega_y \sin \psi) \cos \frac{\theta}{2} \sin \frac{\phi - \psi}{2} + [(\omega_x \sin \psi + \omega_y \cos \psi) / \sin \theta - \{\omega_z - (\omega_x \sin \psi + \omega_y \cos \psi) \frac{\cos \theta}{\sin \theta}\}] \sin \frac{\theta}{2} \cos \frac{\phi - \psi}{2}$$

$$= \begin{cases} \omega_x : \cos \psi \cos \frac{\theta}{2} \sin \frac{\phi - \psi}{2} + \sin \psi (1 + \cos \theta) / \sin \theta \times \sin \frac{\theta}{2} \cos \frac{\phi - \psi}{2} \\ \omega_y : -\sin \psi \cos \frac{\theta}{2} \sin \frac{\phi - \psi}{2} + \cos \psi (1 + \cos \theta) / \sin \theta \times \sin \frac{\theta}{2} \cos \frac{\phi - \psi}{2} \\ \omega_z : -\sin \frac{\theta}{2} \cos \frac{\phi - \psi}{2} = -\eta \quad \leftarrow \text{minus } -\eta \end{cases}$$

$$\begin{aligned}
2\dot{\eta} &= 2 \left(\sin \frac{\theta}{2} \cos \frac{\phi - \psi}{2} \right)' \\
&= \dot{\theta} \cos \frac{\theta}{2} \cos \frac{\phi - \psi}{2} - (\dot{\phi} - \dot{\psi}) \sin \frac{\theta}{2} \sin \frac{\phi - \psi}{2} \\
&= (\omega_x \cos \psi - \omega_y \sin \psi) \cos \frac{\theta}{2} \cos \frac{\phi - \psi}{2} + [-(\omega_x \sin \psi + \omega_y \cos \psi) / \sin \theta \\
&\quad + [\omega_z - (\omega_x \sin \psi + \omega_y \cos \psi) \cos \theta] / \sin \theta] \sin \frac{\theta}{2} \sin \frac{\phi - \psi}{2}
\end{aligned}$$

$$= \begin{cases} \omega_x : \cos \psi \cos \frac{\theta}{2} \cos \frac{\phi - \psi}{2} - \sin \psi (1 - \cos \theta) / \sin \theta \times \sin \frac{\theta}{2} \sin \frac{\phi - \psi}{2} \\ \omega_y : -\sin \psi \cos \frac{\theta}{2} \cos \frac{\phi - \psi}{2} - \cos \psi (1 - \cos \theta) / \sin \theta \times \sin \frac{\theta}{2} \sin \frac{\phi - \psi}{2} \\ \omega_z : \sin \frac{\theta}{2} \sin \frac{\phi - \psi}{2} = \xi \quad \leftarrow \text{minus} - \xi \end{cases}$$

$$\begin{aligned}
2\dot{\zeta} &= 2 \left(\cos \frac{\theta}{2} \sin \frac{\phi + \psi}{2} \right)' \\
&= -\dot{\theta} \sin \frac{\theta}{2} \sin \frac{\phi + \psi}{2} + (\dot{\phi} + \dot{\psi}) \cos \frac{\theta}{2} \cos \frac{\phi + \psi}{2} \\
&= -(\omega_x \cos \psi - \omega_y \sin \psi) \sin \frac{\theta}{2} \sin \frac{\phi + \psi}{2} + [(\omega_x \sin \psi + \omega_y \cos \psi) / \sin \theta \\
&\quad + \omega_z - (\omega_x \sin \psi + \omega_y \cos \psi) \cos \theta / \sin \theta] \cos \frac{\theta}{2} \cos \frac{\phi + \psi}{2}
\end{aligned}$$

$$= \begin{cases} \omega_x : -\cos \psi \sin \frac{\theta}{2} \sin \frac{\phi + \psi}{2} + \sin \psi (1 + \cos \theta) / \sin \theta \times \cos \frac{\theta}{2} \cos \frac{\phi + \psi}{2} \\ \omega_y : \sin \psi \sin \frac{\theta}{2} \sin \frac{\phi + \psi}{2} + \cos \psi (1 + \cos \theta) / \sin \theta \times \cos \frac{\theta}{2} \cos \frac{\phi + \psi}{2} \\ \omega_z : \cos \frac{\theta}{2} \cos \frac{\phi + \psi}{2} = \chi \end{cases}$$

$$\begin{aligned}
2\dot{\chi} &= 2\left(\cos\frac{\theta}{2}\cos\frac{\phi+\psi}{2}\right)' = -\dot{\theta}\sin\frac{\theta}{2}\cos\frac{\phi+\psi}{2} - (\dot{\phi} + \dot{\psi})\cos\frac{\theta}{2}\sin\frac{\phi+\psi}{2} \\
&= -(\omega_x \cos\psi - \omega_y \sin\psi)\sin\frac{\theta}{2}\cos\frac{\phi+\psi}{2} + [-(\omega_x \cos\psi + \omega_y \sin\psi)]/\sin\theta \\
&\quad - (\omega_z - (\omega_x \cos\psi + \omega_y \sin\psi)\frac{\cos\theta}{\sin\theta})]\cos\frac{\theta}{2}\sin\frac{\phi+\psi}{2}
\end{aligned}$$

$$= \begin{cases} \omega_x : -\cos\psi \sin\frac{\theta}{2}\cos\frac{\phi+\psi}{2} - \cos\psi(1 + \cos\theta)/\sin\theta \times \cos\frac{\theta}{2}\sin\frac{\phi+\psi}{2} \\ \omega_y : \sin\psi \sin\frac{\theta}{2}\cos\frac{\phi+\psi}{2} - \sin\psi(1 + \cos\theta)/\sin\theta \times \cos\frac{\theta}{2}\sin\frac{\phi+\psi}{2} \\ \omega_z : -\cos\frac{\theta}{2}\sin\frac{\phi+\psi}{2} \end{cases}$$

Check quaternions of A_{ij} coefficients

$$\begin{aligned} A_{11} &= (-\xi^2 + \eta^2 - \zeta^2 + \chi^2) = -\sin^2 \frac{\theta}{2} \sin^2 \frac{\psi - \phi}{2} + \sin^2 \frac{\theta}{2} \cos^2 \frac{\psi - \phi}{2} \\ &\quad - \cos^2 \frac{\theta}{2} \sin^2 \frac{\psi + \phi}{2} + \cos^2 \frac{\theta}{2} \cos^2 \frac{\psi + \phi}{2} \\ &= \sin^2 \frac{\theta}{2} (-\sin^2 \frac{\psi - \phi}{2} + \cos^2 \frac{\psi - \phi}{2}) - \cos^2 \frac{\theta}{2} (\sin^2 \frac{\psi + \phi}{2} - \cos^2 \frac{\psi + \phi}{2}) \\ &= \sin^2 \frac{\theta}{2} (2 \cos^2 \frac{\psi - \phi}{2} - 1) + \cos^2 \frac{\theta}{2} (\cos^2 \frac{\psi + \phi}{2} - 1) \\ &= \sin^2 \frac{\theta}{2} (1 + \cos(\psi - \phi) - 1) + \cos^2 \frac{\theta}{2} (1 + \cos(\psi + \phi) - 1) \\ &= \sin^2 \frac{\theta}{2} \cos(\psi - \phi) + \cos^2 \frac{\theta}{2} \cos(\psi + \phi) \\ &= \sin^2 \frac{\theta}{2} (\cos \psi \cos \phi + \sin \psi \sin \phi) + \cos^2 \frac{\theta}{2} (\cos \psi \cos \phi - \sin \psi \sin \phi) \\ &= \frac{1}{2} (1 - \cos \theta) (\cos \psi \cos \phi + \sin \psi \sin \phi) + \frac{1}{2} (1 + \cos \theta) (\cos \psi \cos \phi - \sin \psi \sin \phi) \\ &= \cos \psi \cos \phi - \cos \theta \sin \psi \sin \phi \end{aligned}$$

$$\begin{aligned}
A_{12} &= 2(\varsigma\chi - \xi\eta) = 2\left(\cos^2 \frac{\theta}{2} \sin \frac{\psi + \phi}{2} \cos \frac{\psi + \phi}{2} - \sin^2 \frac{\theta}{2} \sin \frac{\psi - \phi}{2} \cos \frac{\psi - \phi}{2}\right) \\
&= (1 + \cos \theta) \sin \frac{\psi + \phi}{2} \cos \frac{\psi + \phi}{2} - (1 - \cos \theta) \sin \frac{\psi - \phi}{2} \cos \frac{\psi - \phi}{2} \\
&= \sin \frac{\psi + \phi}{2} \cos \frac{\psi + \phi}{2} - \sin \frac{\psi - \phi}{2} \cos \frac{\psi - \phi}{2} + \cos \theta \left(\sin \frac{\psi + \phi}{2} \cos \frac{\psi + \phi}{2} \right. \\
&\quad \left. + \sin \frac{\psi - \phi}{2} \cos \frac{\psi - \phi}{2} \right) \\
&= \frac{1}{2} \sin\left(\frac{\psi + \phi}{2} + \frac{\psi + \phi}{2}\right) - \frac{1}{2} \sin\left(\frac{\psi - \phi}{2} + \frac{\psi - \phi}{2}\right) + \frac{1}{2} \cos \theta (\sin(\psi + \phi) \\
&\quad + \sin(\psi - \phi)) \\
&= \frac{1}{2} \sin(\psi + \phi) - \frac{1}{2} \sin(\psi - \phi) + \frac{1}{2} \cos \theta (\sin(\psi + \phi) - \frac{1}{2} \sin(\psi - \phi)) \\
&= \frac{1}{2} (\sin \psi \cos \phi + \cos \psi \sin \phi - \sin \psi \cos \phi + \cos \psi \sin \phi) + \frac{1}{2} \cos \theta (\text{same}) \\
&= \cos \psi \sin \phi + \cos \theta \sin \psi \cos \phi
\end{aligned}$$

$$\begin{aligned}
A_{13} &= 2(\eta\zeta + \xi\chi) = 2\left(\sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{\psi-\phi}{2}\sin\frac{\psi+\phi}{2}\right. \\
&\quad \left. + \sin\frac{\theta}{2}\cos\frac{\theta}{2}\sin\frac{\psi-\phi}{2}\cos\frac{\psi+\phi}{2}\right) \\
&= 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\left[\cos\frac{\psi-\phi}{2}\sin\frac{\psi+\phi}{2} + \sin\frac{\psi-\phi}{2}\cos\frac{\psi+\phi}{2}\right] \\
&= \sin\theta\left[\sin\frac{\psi+\phi}{2}\cos\frac{\psi-\phi}{2} + \cos\frac{\psi+\phi}{2}\sin\frac{\psi-\phi}{2}\right] \\
&= \sin\theta\left[\sin\left(\frac{\psi+\phi}{2} + \frac{\psi-\phi}{2}\right)\right] \\
&= \sin\theta\sin\psi
\end{aligned}$$

$$\begin{aligned}
A_{21} &= -2(\xi\eta + \zeta\chi) = -2\left(\sin^2 \frac{\theta}{2} \cos \frac{\psi - \phi}{2} \sin \frac{\psi - \phi}{2} \right. \\
&\quad \left. + \cos^2 \frac{\theta}{2} \sin \frac{\psi + \phi}{2} \cos \frac{\psi + \phi}{2} \right) \\
&= -(1 - \cos \theta) \cos \frac{\psi - \phi}{2} \sin \frac{\psi - \phi}{2} - (1 + \cos \theta) \sin \frac{\psi + \phi}{2} \cos \frac{\psi + \phi}{2} \\
&= -\frac{1}{2}(1 - \cos \theta) \sin(\psi - \phi) - \frac{1}{2}(1 + \cos \theta) \sin(\psi + \phi) \\
&= -\frac{1}{2}(\sin(\psi - \phi) + \sin(\psi + \phi)) + \frac{1}{2} \cos \theta (\sin(\psi - \phi) - \sin(\psi + \phi)) \\
&= -\frac{1}{2}(sc - cs + sc + cs) + \frac{1}{2} \cos \theta (sc - cs - sc - cs) \\
&= -\sin \psi \cos \phi - \cos \theta \cos \psi \sin \phi
\end{aligned}$$

$$\begin{aligned}
A_{22} &= \xi^2 - \eta^2 - \zeta^2 + \chi^2 \\
&= \sin^2 \frac{\theta}{2} \left(\sin^2 \frac{\psi - \phi}{2} - \cos^2 \frac{\psi - \phi}{2} \right) + \cos^2 \frac{\theta}{2} \left(-\sin^2 \frac{\psi + \phi}{2} + \cos^2 \frac{\psi + \phi}{2} \right) \\
&= \sin^2 \frac{\theta}{2} \left(1 - 2 \cos^2 \frac{\psi - \phi}{2} \right) + \cos^2 \frac{\theta}{2} \left(2 \cos^2 \frac{\psi + \phi}{2} - 1 \right) \\
&= -\sin^2 \frac{\theta}{2} \cos(\psi - \phi) + \cos^2 \frac{\theta}{2} \cos(\psi + \phi) \\
&= -\frac{1}{2} (1 - \cos \theta) \cos(\psi - \phi) + \frac{1}{2} (1 + \cos \theta) \cos(\psi + \phi) \\
&= \frac{1}{2} (-\cos(\psi - \phi) + \cos(\psi + \phi)) + \frac{1}{2} \cos \theta (\cos(\psi - \phi) + \cos(\psi + \phi)) \\
&= \frac{1}{2} (-cc - ss + cc - ss) + \frac{1}{2} \cos \theta (cc + ss + cc - ss) \\
&= -\sin \psi \sin \phi + \cos \theta \cos \psi \cos \phi
\end{aligned}$$

$$\begin{aligned}
A_{23} &= 2(\eta\chi - \xi\zeta) = 2\left(\sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{\psi+\phi}{2}\cos\frac{\psi-\phi}{2}\right. \\
&\quad \left.- \sin\frac{\theta}{2}\cos\frac{\theta}{2}\sin\frac{\psi+\phi}{2}\sin\frac{\psi-\phi}{2}\right) \\
&= 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\left(\cos\frac{\psi+\phi}{2}\cos\frac{\psi-\phi}{2} - \sin\frac{\psi+\phi}{2}\sin\frac{\psi-\phi}{2}\right) \\
&= 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\left(\frac{\psi+\phi}{2} + \frac{\psi-\phi}{2}\right) \\
&= \sin\theta\cos\psi
\end{aligned}$$

$$\begin{aligned}
A_{31} &= 2(\eta\zeta - \xi\chi) = 2\left(\sin\frac{\theta}{2}\cos\frac{\theta}{2}\sin\frac{\psi+\phi}{2}\cos\frac{\psi-\phi}{2}\right. \\
&\quad \left.- \sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{\psi+\phi}{2}\sin\frac{\psi-\phi}{2}\right) \\
&= 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\left(\sin\frac{\psi+\phi}{2}\cos\frac{\psi-\phi}{2} - \cos\frac{\psi+\phi}{2}\sin\frac{\psi-\phi}{2}\right) \\
&= 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\sin\left(\frac{\psi+\phi}{2} - \frac{\psi-\phi}{2}\right) \\
&= \sin\theta\sin\phi
\end{aligned}$$

$$\begin{aligned}
A_{32} &= -2(\xi\zeta + \eta\chi) = -2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\left(\sin\frac{\psi+\phi}{2}\sin\frac{\psi-\phi}{2}\right. \\
&\quad \left.+ \cos\frac{\psi+\phi}{2}\cos\frac{\psi-\phi}{2}\right) \\
&= -\sin\theta\cos\left(\frac{\psi+\phi}{2} - \frac{\psi-\phi}{2}\right) \\
&= -\sin\theta\cos\phi
\end{aligned}$$

$$\begin{aligned}
A_{33} &= -\xi^2 - \eta^2 + \varsigma^2 + \chi^2 \\
&= -\sin^2 \frac{\theta}{2} \left\{ \sin^2(-) + \cos^2(-) \right\} + \cos^2 \frac{\theta}{2} \left\{ \sin^2(+) + \cos^2(+) \right\} \\
&= -\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} = \frac{1}{2} \{ -(1 - \cos \theta) + 1 + \cos \theta \} = \cos \theta \quad QED!
\end{aligned}$$

The Lennard-Jones Potentials

With the Coulombic interactions, the 12-6 Lennard-Jones potential is adopted for the TIP4P and TIP5P cases:

$$\Phi(r) = A / r^{12} - B / r^6$$

for TIP4 :

$$A = 4.17 \times 10^{-8} \text{ erg} \cdot \text{Ang}^{12}, B = 4.24 \times 10^{-11} \text{ erg} \cdot \text{Ang}^6$$

for TIP5 – Ewald sum :

$$A = 3.85 \times 10^{-8} \text{ erg} \cdot \text{Ang}^{12}, B = 4.36 \times 10^{-11} \text{ erg} \cdot \text{Ang}^6$$

Some parameters are,

$$r(\text{OH}) = 0.9572 \text{ Ang}, \Delta\text{HOH} = 104.52^\circ$$

$$r(\text{OM}) = 0.15 \text{ Ang for TIP4P only}$$

The equipartition line of the virtual M site is on the plain that equally separates the HOH angle. The TIP5P cases are also available.

To Start a Run

To start a simulation of water cluster with the TIP5P code, the adjacent 4x4 hydrogen pairs are summed electrostatically, and oxygen pairs are coupled by TIP5P Lennard-Jones potentials.

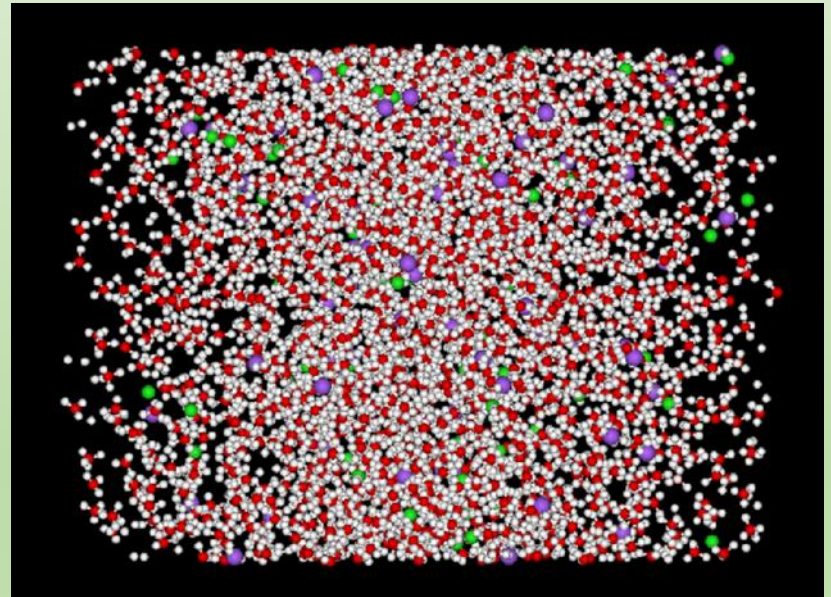
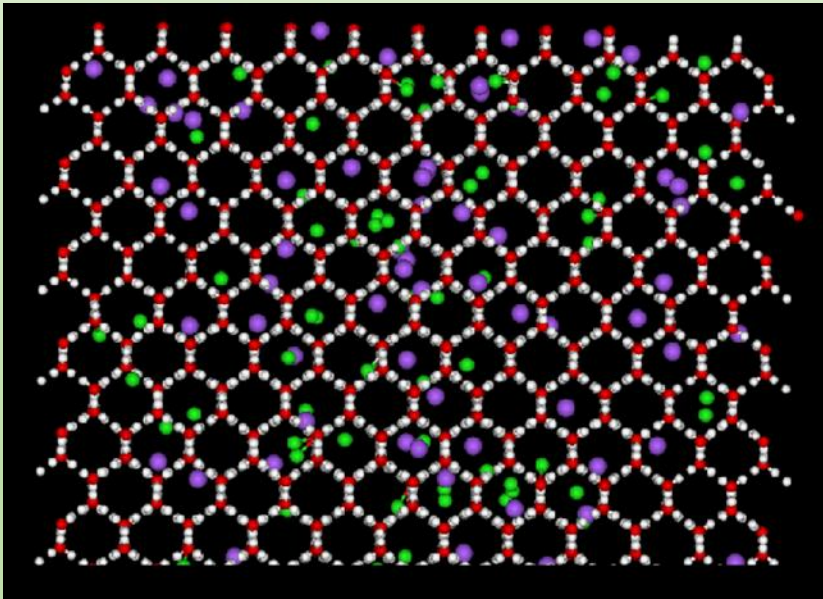
To get an initial state, we make the size of at least a 6x6x6 water cluster for numerical stability. Short-range and long-range Coulombic forces are best separated for interactions, and the short-range forces are made to be spatially dumped.

Around a given temperature, a dryrun is executed at least for 5 periods that is 10^{-9} seconds. The long dryrun is very important !!

Then, we apply the electric field $E_x = E_0 \sin(\omega \cdot \text{time})$ in the x-direction to excite the electric dipole interactions of water. For the moment, we give the electric field 10 GHz where the electric field E_0 and electric dipole p_0 are of the order of 5×10^{-3} eV.

To Obtain the Initial Equilibrium for 298 K

We will use salt ions of Na(+) and Cl(-) initially as the dryrun to give random noises. The 6x6x6 water clusters have 64 Na(+) and 64 Cl(-) ions, and a run time is $t=1,700$. We can see random water clusters. Afterwards, the salt ions are gradually removed, and the dryrun is continued for 5 periods up to $t=50,000$.



At start and the end of the dryrun of salt ions of $t=0 - 3700$. The dryrun without ions is continued afterward for 5 periods.